



Itô Calculus Applied to SDEs in the Black-Scholes Model

Keshav Ramesh | DRP Spring 2025

Why Stochastic Models for Asset Prices?



- Financial markets are inherently uncertain and fluctuate continuously.
- Prices respond to unpredictable news, sentiment, and events.
- To model this, we use stochastic processes — especially Brownian motion — to capture randomness in continuous time.

Brownian Motion: The Core Model of Randomness



Let B_t be standard Brownian motion. It satisfies:

- $B_0 = 0$
- Independent, normally distributed increments: $B_t - B_s \sim \mathcal{N}(0, t - s)$
- Continuous but nowhere differentiable paths.
- $E[B_t] = 0, \text{Var}(B_t) = t$

Why Classical Calculus Fails



- Brownian motion has infinite total variation and is nowhere differentiable.
- Riemann–Stieltjes integrals require bounded variation $\rightarrow$ fail for B_t
- Example:

$$\lim_{\|P\| \rightarrow 0} \sum f(t_i) (B_{t_{i+1}} - B_{t_i})$$

doesn't converge like a Riemann sum.

- We need a new definition of integration: **the Itô integral**.

The Itô Integral: A New Kind of Integration

- For adapted process H_t , define:

$$\int_0^T H_t dB_t := \lim_{\|P\| \rightarrow 0} \sum H_{t_i} (B_{t_{i+1}} - B_{t_i})$$

with H_t , non-anticipating.

- Properties:

- Linearity:

$$\int (aH + bG) dB = a \int H dB + b \int G dB$$

- Martingale:

$$E \left[\int_0^T H_t dB_t \middle| \mathcal{F}_s \right] = \int_0^s H_t dB_t$$

- Isometry:

$$E \left[\left(\int_0^T H_t dB_t \right)^2 \right] = E \left[\int_0^T H_t^2 dt \right]$$

Itô's Formula (Basic Version)



For $f \in C^2(R)$ and B_t standard Brownian motion:

$$df(B_t) = f'(B_t) dB_t + \frac{1}{2} f''(B_t) dt$$

is equivalent to the stochastic version of the Chain Rule.

It captures both:

- Random fluctuations via dB_t
- Accumulated variance via B_t

Itô's Formula (General Form)

Let X_t be an Itô process:

$$dX_t = \mu_t dt + \sigma_t dB_t$$

Then for $f(t, X_t) \in C^{1,2}$, we have:

$$df(t, X_t) = f_t dt + f_x dX_t + \frac{1}{2} f_{xx} (dX_t)^2$$

Substituting dX_t , we have that:

$$df = \left(f_t + \mu_t f_x + \frac{1}{2} \sigma_t^2 f_{xx} \right) dt + \sigma_t f_x dB_t$$

Enter Stochastic Differential Equations (SDEs)



A stochastic differential equation models the evolution of a random process:

$$dX_t = \mu(t, X_t) dt + \sigma(t, X_t) dB_t$$

Where:

- $\mu(t, X_t)$: drift term (represents deterministic trend)
- $\sigma(t, X_t)$: diffusion term (represents random fluctuation)

Geometric Brownian Motion (GBM)



Geometric Brownian motion is the standard model for stock prices:

$$dS_t = \mu S_t dt + \sigma S_t dB_t$$

Solution using Itô's formula:

$$S_t = S_0 \exp \left[\left(\mu - \frac{1}{2} \sigma^2 \right) t + \sigma B_t \right]$$

Properties:

- $S_t > 0$ for all t
- Log-normal distribution

Goal: Price a Call Option



A European call option gives the right to buy a stock at strike price K at time T :

$$\text{Payoff} = \max(S_T - K, 0)$$

The objective is to find the fair price $V(t, S_t)$ at current time $t < T$.

- Need to use the SDE for the stock, Itô's formula, and the no-arbitrage argument to derive a PDE.

Replication: The Pricing Principle



Build a self-financing portfolio (where no money is added or withdrawn before maturity), as follows:

$$V_t = a_t S_t + b_t \beta_t$$

- S_t : stock (risky asset)
- $\beta_t = e^{rt}$: bond (risk-free asset)

No-arbitrage principle: if the portfolio perfectly replicates the option payoff, then:

- $V_t =$ option price $\rightarrow$ same value

Deriving the Black-Scholes PDE

Assume $V(t, S_t)$ is the option price.

Apply Itô's formula, as follows:

$$dV = \left(V_t + \mu S V_S + \frac{1}{2} \sigma^2 S^2 V_{SS} \right) dt + \sigma S V_S dB_t$$

The replicating portfolio also satisfies the following:

$$dV = a_t dS_t + b_t d\beta_t$$

Then, we match terms $\rightarrow$ eliminate $dB_t \rightarrow$ get the resulting PDE:

$$\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + r S \frac{\partial V}{\partial S} - r V = 0$$

Solving the PDE: Black-Scholes Formula



The solution to the PDE for a European call option is as follows:

$$C(S, t) = S\Phi(d_1) - Ke^{-r(T-t)}\Phi(d_2)$$

where:

$$d_1 = \frac{\log\left(\frac{S}{K}\right) + \left(r + \frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}}, \quad d_2 = d_1 - \sigma\sqrt{T-t}$$

Note that $\Phi(\cdot)$ refers to the cumulative distribution function of the standard normal distribution.

Why Itô Calculus Matters



As we just saw, Itô calculus enabled:

- Rigorous integration with Brownian motion
- Construction and solution of SDEs for asset prices
- Derivation of the Black-Scholes PDE
- Risk-neutral pricing through martingale methods

Itô calculus connects randomness, differential equations, and no-arbitrage pricing.

Conclusion



To recap:

- Brownian motion $\rightarrow$ Itô integral $\rightarrow$ SDEs $\rightarrow$ Itô's formula
- Derived Black-Scholes PDE and solved it using replication
- Showed how randomness and pricing are unified

The main takeaway: **Itô calculus makes modern derivative pricing both possible and precise.**



Thank you!

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References



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